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CALIBRATION REPORT

COORDINATE MEASURING MACHINE

CVA800A



Certificate No: VNC-81-0749

Decree No.105: DK 522

CERTIFICATE OF CALIBRATION

Customer: CÔNG TY TNHH CÔNG NGHỆ MÁY VĂN PHÒNG KYOCERA VIỆT NAM
Place of Calibration: LÔ ĐẤT SỐ 56A, 56B VÀ 56C, KHU ĐÔ THỊ, CÔNG NGHIỆP VÀ DỊCH VỤ, PHƯỜNG HÒA BÌNH, TP HẢI PHÒNG, VIỆT NAM
Equipment Name: COORDINATE MEASURING MACHINE
Type: CVA800A
Manufacturer: TOKYO SEIMITSU CO.,LTD
Manufacturing Date: Jul-2012
Serial Number: GB6252HG
Calibrated by: CONG NAM
Calibration Date: May 20, 2026
Temperature: 22.0 °C **Humidity:** 56 %
Method of Calibration: QT.HC-03
Standard used: The calibration was carried out by using the following instruments:

Instruments	Specification	Control No.	Calibration Validity	Calibration Lab.
GAUGE BLOCK	25mm - 1000mm	ACCTV-034	Nov-2026	JQA Japan

Result of Calibration & Uncertainty of Measurement: The result and uncertainty of this calibration are shown on the attached report of calibration.

We certify that the above product was calibrated based on the authorized standard calibration procedure of ACCRETECH VIETNAM CO., LTD

The measuring instruments are traceable to the national measurement standards certified by National Institute of Advanced Industrial Science and Technology of Japan.

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Date of issue: May 22, 2026

Calibration Laboratory

ACCRETECH VIETNAM CO.,LTD



CALIBRATION REPORT

A. CALIBRATION RESULT

[The temperature condition A]

$$E_{0,MPE} = 2.9 + 5.0L/1000 \mu m$$

L=Meas.Length (Unit: mm)/ Result (Unit: μm)

No	Item	Result					
1	E0 Length measurement error X-axis	Length (mm)	50	100	200	400	600
		Mean Error (μm)	-0.47	0.63	0.97	1.54	0.16
		Repeatability (μm)	0.2	0.3	0.5	0.4	0.4
2	E0 Length measurement error Y-axis	Length (mm)	50	100	200	300	500
		Mean Error (μm)	0.41	-0.51	0.71	-0.74	0.84
		Repeatability (μm)	0.3	0.5	0.4	0.3	0.4
3	E0 Length measurement error Z-axis	Length (mm)	50	100	200	300	500
		Mean Error (μm)	-0.83	0.69	-1.13	1.08	-0.32
		Repeatability (μm)	0.2	0.1	0.2	0.2	0.1
4	E0 Length measurement error Spatial (position 1)	Length (mm)	50	100	200	400	600
		Mean Error (μm)	-0.05	-0.99	-2.13	-2.14	1.34
		Repeatability (μm)	0.5	0.5	0.7	0.7	0.3
5	E0 Length measurement error Spatial (position 2)	Length (mm)	50	100	200	400	600
		Mean Error (μm)	-0.83	0.87	-1.47	1.46	1.76
		Repeatability (μm)	0.2	0.3	0.5	0.5	0.7
6	E0 Length measurement error Spatial (position 3)	Length (mm)	50	100	200	400	600
		Mean Error (μm)	-0.69	-0.85	0.25	1.64	1.46
		Repeatability (μm)	0.5	0.2	0.1	0.7	0.2
7	E0 Length measurement error Spatial (position 4)	Length (mm)	50	100	200	400	600
		Mean Error (μm)	0.45	-0.47	-1.03	-0.86	0.14
		Repeatability (μm)	0.3	0.3	0.5	0.5	0.6



CALIBRATION REPORT

No	Item	Reference Tolerance	Result	Note
8	Parallelism for XY movement of granite table	20 μm	9.2 μm	
9	Probing error	TP200 3.0 μm	2.3 μm	MPE_P

Estimated uncertainty of measurement is $U = 1.4 \mu\text{m}$ with $k=2$, level of Confidence Approximately 95%).

B. FUNCTIONAL INSPECTION

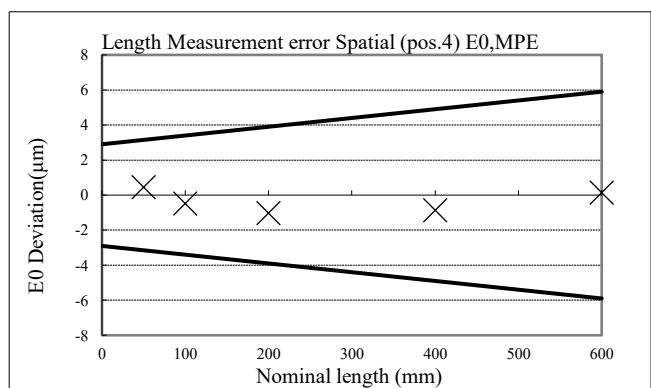
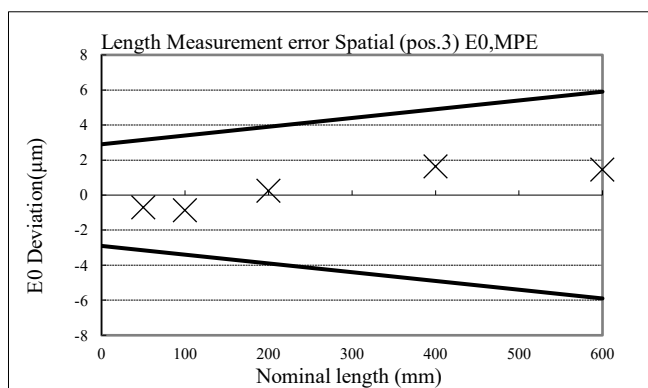
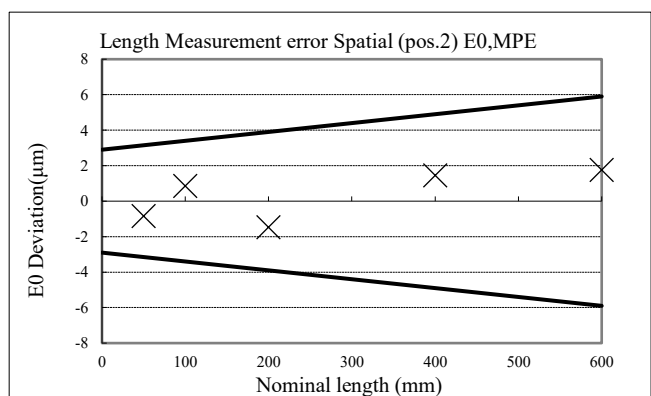
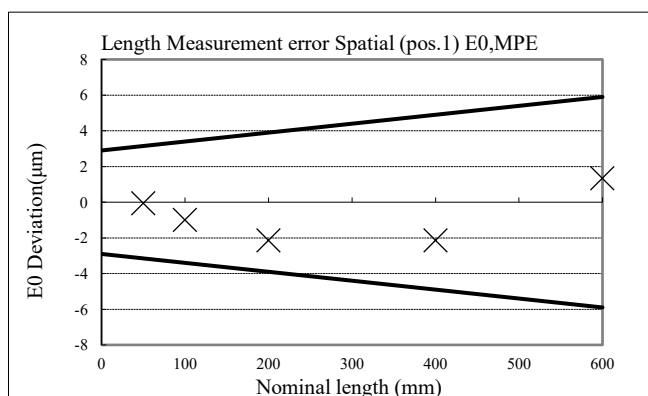
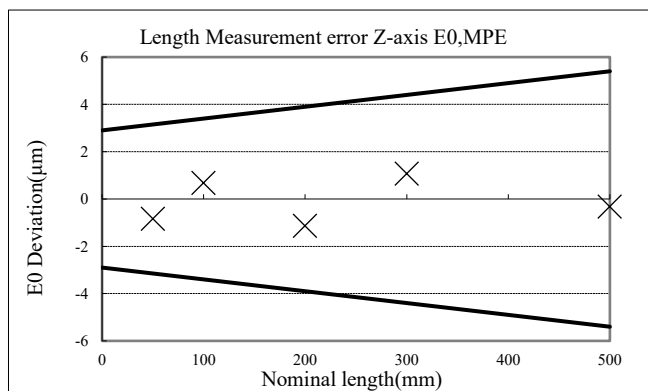
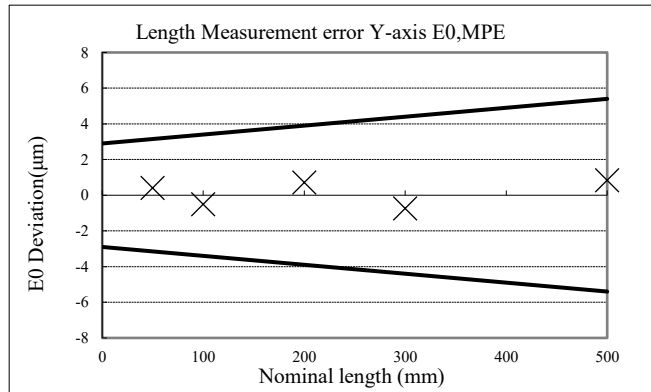
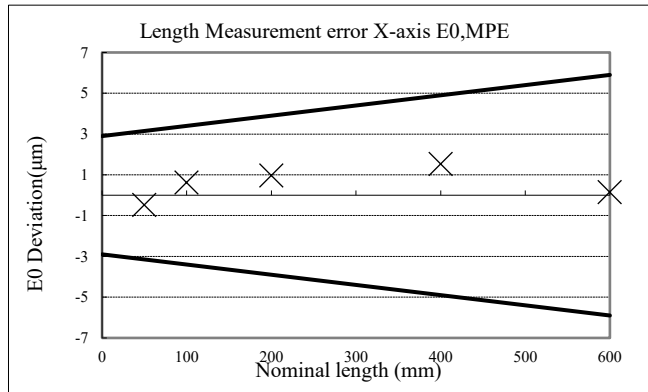
No	Item	Contents	Result	Note
1	MECHANICAL FUNCTION	1) Smoothness of operation of each axis	OK	
		2) Operation distance of each axis		
		3) Clamping power of each axis		
		4) Function of air pressure switch		
		5) Stone level for setting		
2	ELECTRICAL FUNCTION	1) Scale reading signal of each axis	OK	
		2) Controller and control panel function		
3	CONTROLLER & SOFTWARE FUNCTION	1) Start of processor	OK	
		2) Execution of software		
		3) Printer function		
4	SAFETY DEVICES FUNCTION	1) Function of emergency stop switch	OK	
		2) Function of Z-axis fall prevention device		
5	ACCESSORIES	1) Probe head behavior	OK	
		2) Option parts and software function		



CALIBRATION REPORT

TYPE : CVA800A

The plot of Length Measurement error (E0,MPE and R0,MPL)





CALIBRATION REPORT

TYPE : CVA800A

Calibration environment condition

The result of calibration of CMM is affected by room environment.

The environment condition of

1. Environment conditions

1-1 The standard temperature is 20 degrees.

The tolerance of inspection report is the value at the temperature with 20 degrees.

1-2 The tolerance of environmental temperature is following.

- 1 Environmental temperature 20 degrees $\pm$ 2 degrees
- 2 Temperature change at a certain point 1.0 degrees/ 8 hour
- 3 Temperature gradient 0.5 degrees/m

The temperature gradient is the one to exclude the range of 0.3 m from the floor level.

2. Floor Vibration

2-1 The tolerance of vibration while the accuracy inspection is as follows.

2-2 If the air anti-vibration device is not installed.

- | | |
|---|---------------------------------------|
| -1 Number of vibration : Under 10 Hz | Max. Amplitude 0.7 μ m |
| -2 Number of vibration : More than 10 Hz, Under 50 Hz | Max. Acceleration 1 mm/S ² |

2-3 If the anti-vibration device is installed

- | | |
|---|---------------------------------------|
| -1 Number of vibration : Under 10 Hz | Max. Amplitude 1.7 μ m |
| -2 Number of vibration : More than 10 Hz, Under 50 Hz | Max. Acceleration 8 mm/S ² |

3. Relative humidity

3-1 The tolerance of relative humidity while the accuracy inspection is as follows.

- 1 Relative humidity is 50 % $\pm$ 10 %.

4-1 The temperature conditions A while the accuracy inspection is as follows.

- 1 Environmental temperature(degrees) 18-22 degrees Temperature change(degrees/hour) 1.0 degrees
- 2 Temperature change(degrees/day) 2.0 degrees Temperature gradient(degrees/m) 1.0 degrees

4-2 The temperature conditions C while the accuracy inspection is as follows.

- 1 Environmental temperature(degrees) 16-26 degrees Temperature change(degrees/hour) 1.0 degrees
- 2 Temperature change($^{\circ}$ C/day) 5.0 $^{\circ}$ C Temperature gradient($^{\circ}$ C/m) 1.0 $^{\circ}$ C

4-3 The temperature conditions E while the accuracy inspection is as follows.

- 1 Environmental temperature($^{\circ}$ C) 15~30 $^{\circ}$ C Temperature change($^{\circ}$ C/hour) 2.0 $^{\circ}$ C
- 2 Temperature change($^{\circ}$ C/day) 5.0 $^{\circ}$ C Temperature gradient($^{\circ}$ C/m) 1.0 $^{\circ}$ C

(Note1) The temperature gradient is the one to exclude the range of 0.3 m from the floor level.

(Note2) Above Environmental is For conditions Customer Side.